

Class 5: Data Viz with ggplot

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Intro to ggplot

There are many graphics systems in R (ways to make plots and figures). These include “base” R plots. Today we will focus mostly on **ggplot2** package.

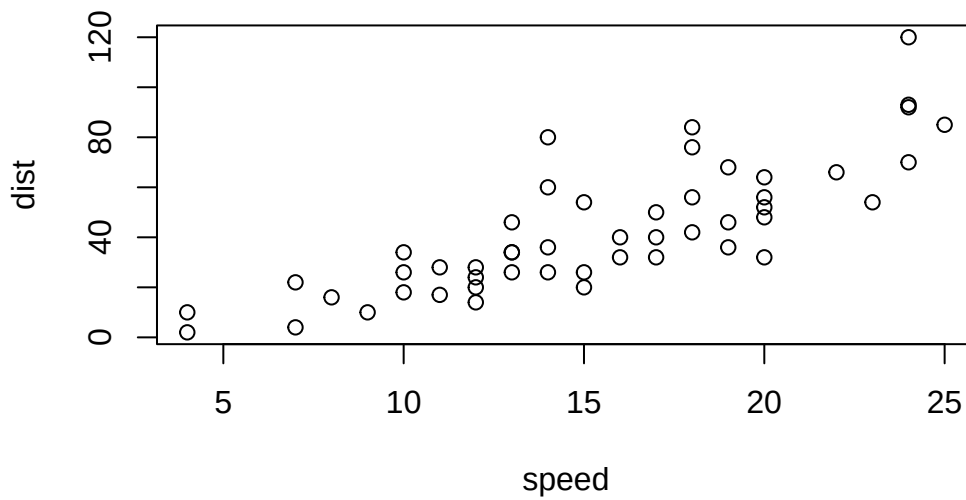
Let’s start with a plot of a simple in-built dataset called **cars**.

```
cars
```

	speed	dist
1	4	2
2	4	10
3	7	4
4	7	22
5	8	16
6	9	10
7	10	18
8	10	26
9	10	34
10	11	17
11	11	28
12	12	14
13	12	20
14	12	24
15	12	28
16	13	26
17	13	34
18	13	34
19	13	46
20	14	26
21	14	36
22	14	60

23	14	80
24	15	20
25	15	26
26	15	54
27	16	32
28	16	40
29	17	32
30	17	40
31	17	50
32	18	42
33	18	56
34	18	76
35	18	84
36	19	36
37	19	46
38	19	68
39	20	32
40	20	48
41	20	52
42	20	56
43	20	64
44	22	66
45	23	54
46	24	70
47	24	92
48	24	93
49	24	120
50	25	85

```
plot(cars)
```

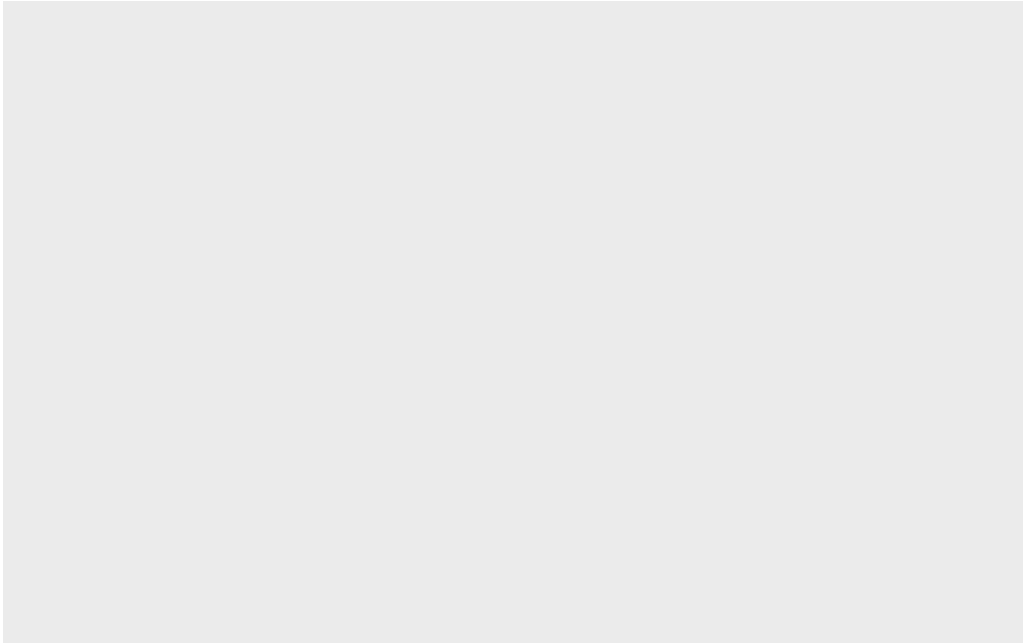


Let's see how we can make this figure using **ggplot**. First I need to install this package on my computer. To install any R package I use the function `install.packages()`

I will run `install.packages("ggplot2")` in my R console not this quarto document!

Before I can use any functions from add on packages I need to load the package from my "library()" with the `library(ggplot2)` call.

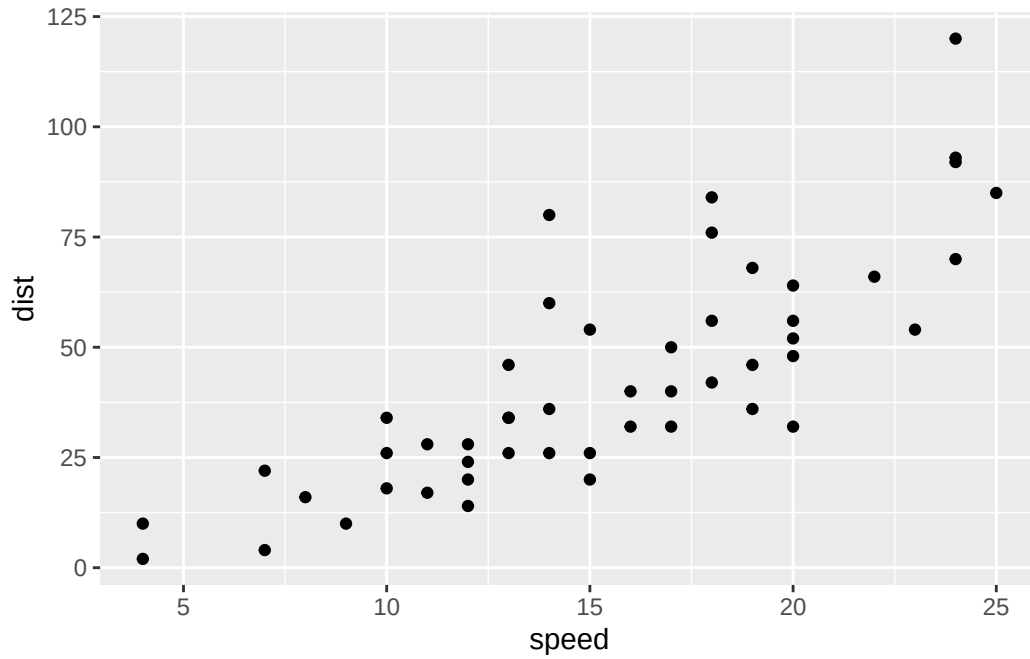
```
#install.packages("ggplot2")  
library(ggplot2)  
ggplot(cars)
```



All ggplot figures have at least 3 things (called layers). These include:

- **data** (the input dataset I want to plot from),
- **aes** (the aesthetic mapping of the data to my plot),
- **geoms** (the `geom_point()`, `geom_line()`, etc. that I want to draw).

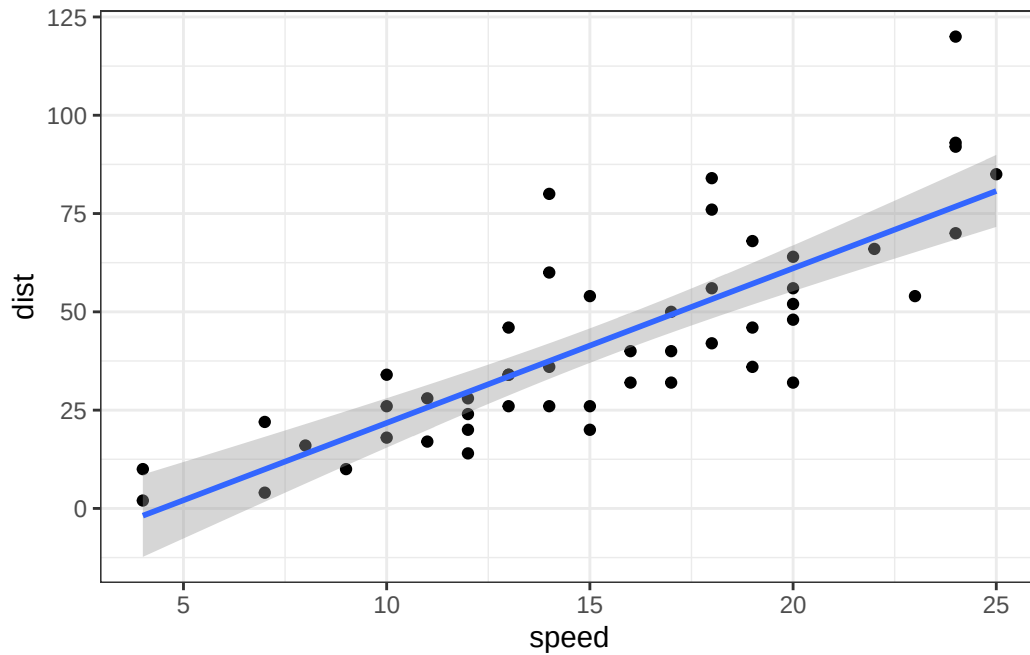
```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point()
```



Let's add a line to show the relationship here:

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method = "lm") +  
  theme_bw()
```

`geom\_smooth()` using formula = 'y ~ x'



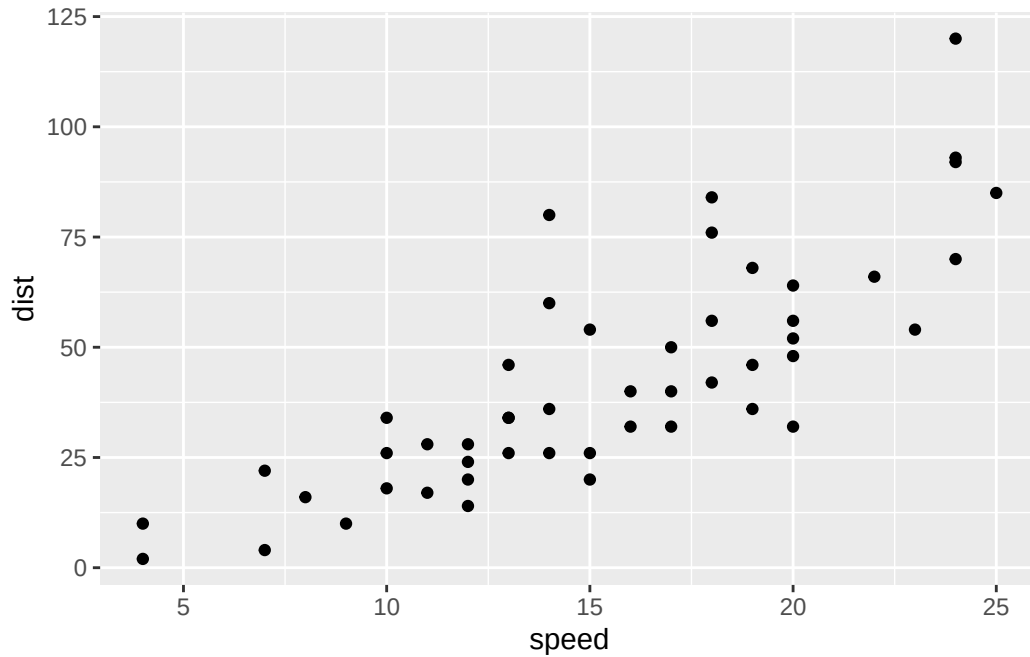
```
labs(title="My first GGPlot")
```

```
$title
[1] "My first GGPlot"

attr("class")
[1] "labels"
```

Q. Which geometric layer should be used to create scatter plots in ggplot2?

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point()
```

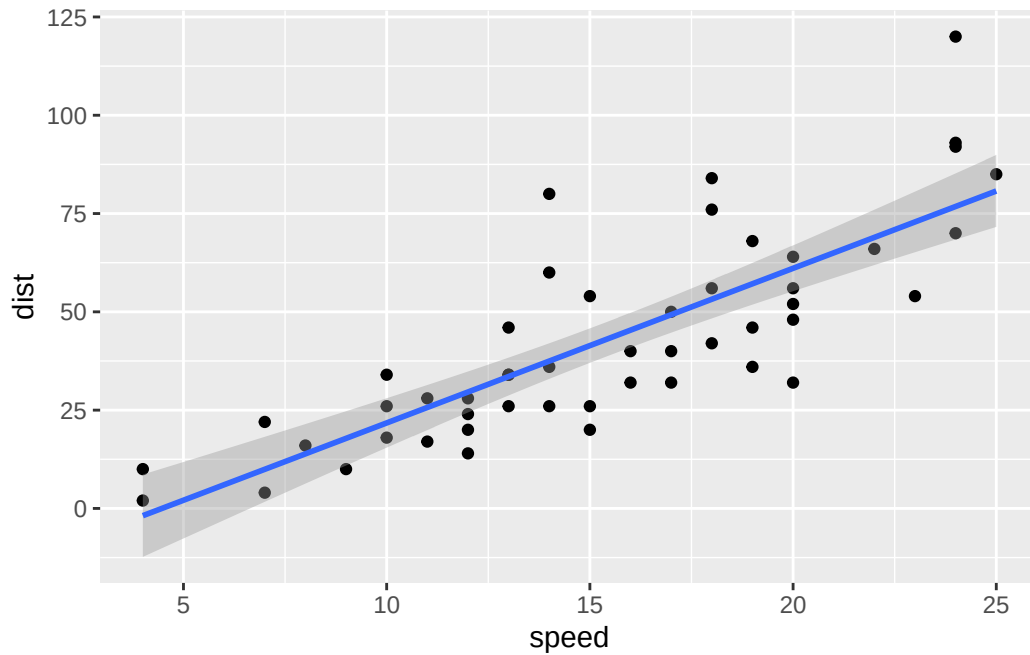


Q. In your own RStudio can you add a trend line layer to help show the relationship between the plot variables with the `geom_smooth()` function?

`geom_smooth()` is used to add a trendline.

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method="lm")
```

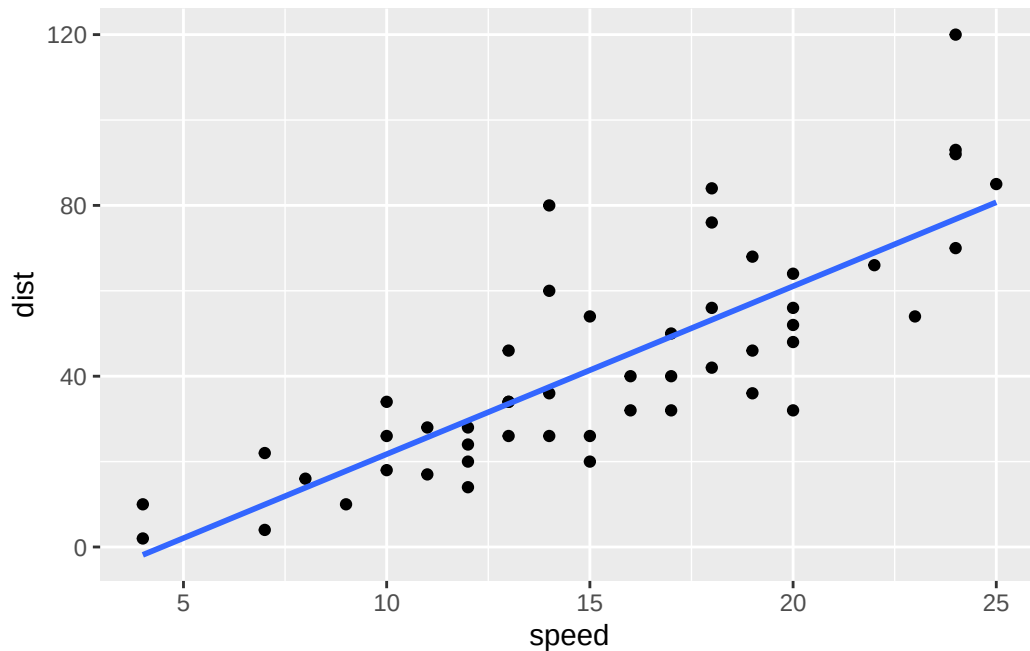
``geom_smooth()`` using formula = 'y ~ x'



Q. Argue with `geom_smooth()` to add a straight line from a linear model without the shaded standard error region?

```
ggplot(cars) +  
  aes(x=speed, y=dist) +  
  geom_point() +  
  geom_smooth(method="lm", se=FALSE)
```

`geom_smooth()` using formula = 'y ~ x'



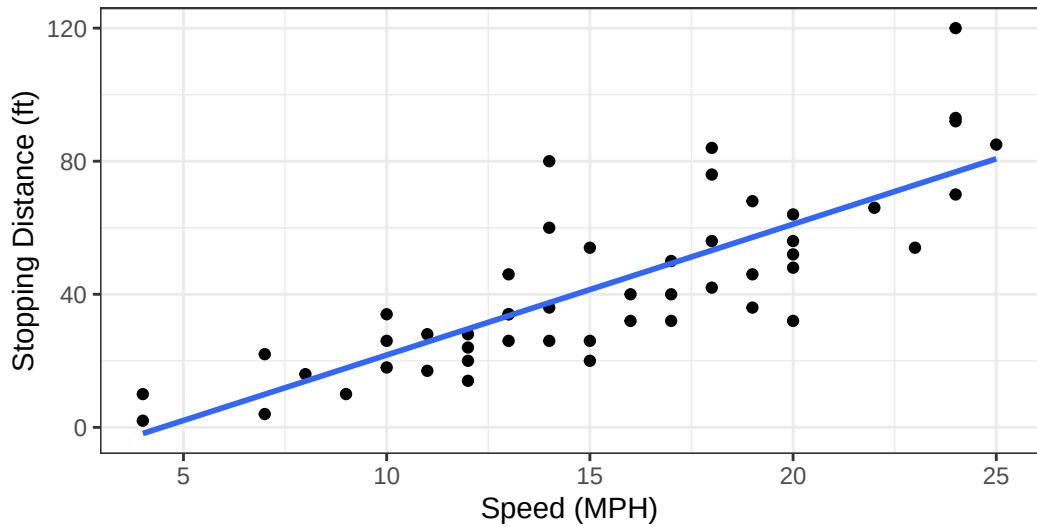
Q. Can you finish this plot by adding various label annotations with the `labs()` function and changing the plot look to a more conservative “black & white” theme by adding the `theme_bw()` function

```
ggplot(cars) +
  aes(x=speed, y=dist) +
  geom_point() +
  labs(title="Speed and Stopping Distances of Cars",
       x="Speed (MPH)",
       y="Stopping Distance (ft)",
       subtitle = "Your informative subtitle text here",
       caption="Dataset: 'cars'") +
  geom_smooth(method="lm", se=FALSE) +
  theme_bw()
```

``geom_smooth()`` using formula = 'y ~ x'

Speed and Stopping Distances of Cars

Your informative subtitle text here



Gene expression figure

The code to read the dataset

```
url <- "https://bioboot.github.io/bimm143_S20/class-material/up_down_expression.txt"
genes <- read.delim(url)
head(genes)
```

	Gene	Condition1	Condition2	State
1	A4GNT	-3.6808610	-3.4401355	unchanging
2	AAAS	4.5479580	4.3864126	unchanging
3	AASDH	3.7190695	3.4787276	unchanging
4	AATF	5.0784720	5.0151916	unchanging
5	AATK	0.4711421	0.5598642	unchanging
6	AB015752.4	-3.6808610	-3.5921390	unchanging

Q2. How many genes are in this dataset?

```
nrow(genes)
```

```
[1] 5196
```

There are 5196 genes.

Q. Use the `colnames()` function and the `ncol()` function on the `genes` data frame to find out what the column names are (we will need these later) and how many columns there are. How many columns did you find?

```
colnames(genes)
```

```
[1] "Gene"      "Condition1" "Condition2" "State"
```

```
ncol(genes)
```

```
[1] 4
```

There are 4 columns.

Q. Use the `table()` function on the `State` column of this `data.frame` to find out how many ‘up’ regulated genes there are. What is your answer?

```
table(genes$State)
```

down	unchanging	up
72	4997	127

127 upregulated genes.

Q. Using your values above and 2 significant figures. What fraction of total genes is up-regulated in this dataset?

```
round( table(genes$State)/nrow(genes), 2)
```

down	unchanging	up
0.01	0.96	0.02

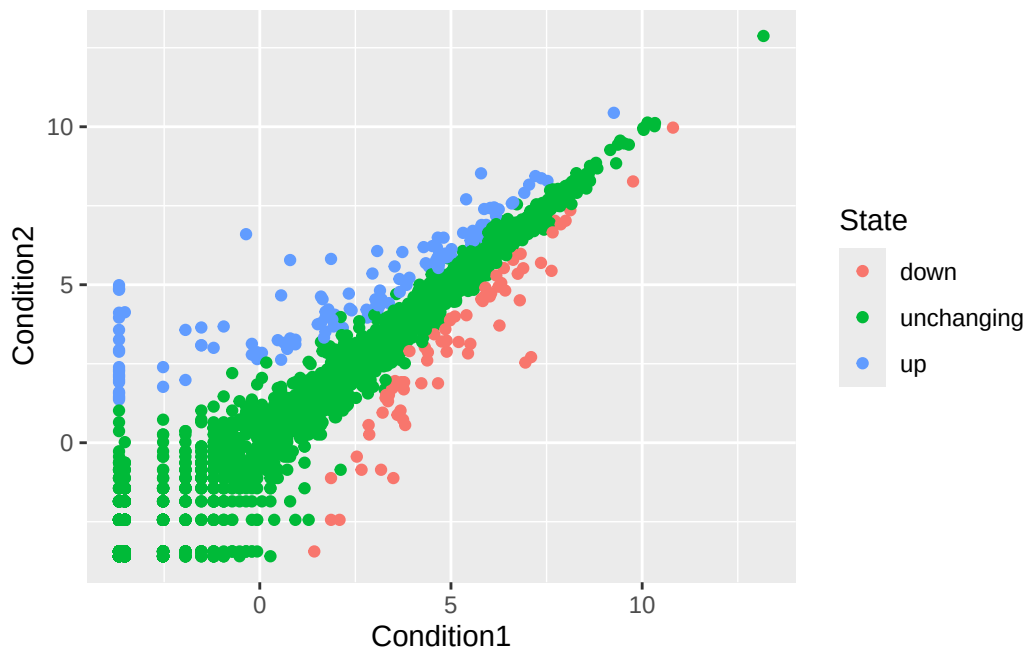
```
n.tot <- nrow(genes)
vals <- table(genes$State)

vals.percent <- vals/n.tot * 100
round(vals.percent, 2)
```

down	unchanging	up
1.39	96.17	2.44

A first plot of this dataset

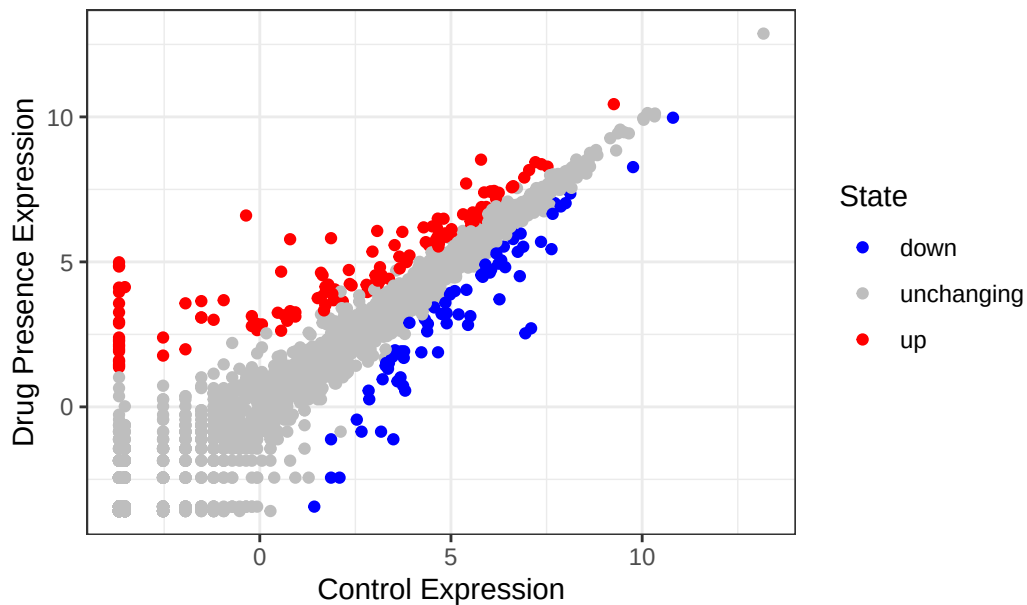
```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point()
```



Adding Aesthetics and colors

```
ggplot(genes) +
  aes(x=Condition1, y=Condition2, col=State) +
  geom_point() +
  theme_bw() +
  labs (title = "Gene Expression changes upon drugs treatment",
        x= "Control Expression",
        y = "Drug Presence Expression") +
  scale_colour_manual (values = c("blue", "gray", "red"))
```

Gene Expression changes upon drugs treatment



7. Going Further

Q. Let's consider the `gapminder_2007` dataset which contains the variables GDP per capita `gdpPercap` and life expectancy `lifeExp` for 142 countries in the year 2007. Complete the code below to produce a first basic scatter plot of this `gapminder_2007` dataset

```
url <- "https://raw.githubusercontent.com/jennybc/gapminder/master/inst/extdata/gapminder.tsv"
gapminder <- read.delim(url)
library(gapminder)
```

Attaching package: 'gapminder'

The following object is masked _by_ '.GlobalEnv':

gapminder

```
#install.packages("dplyr")
library(dplyr)
```

Attaching package: 'dplyr'

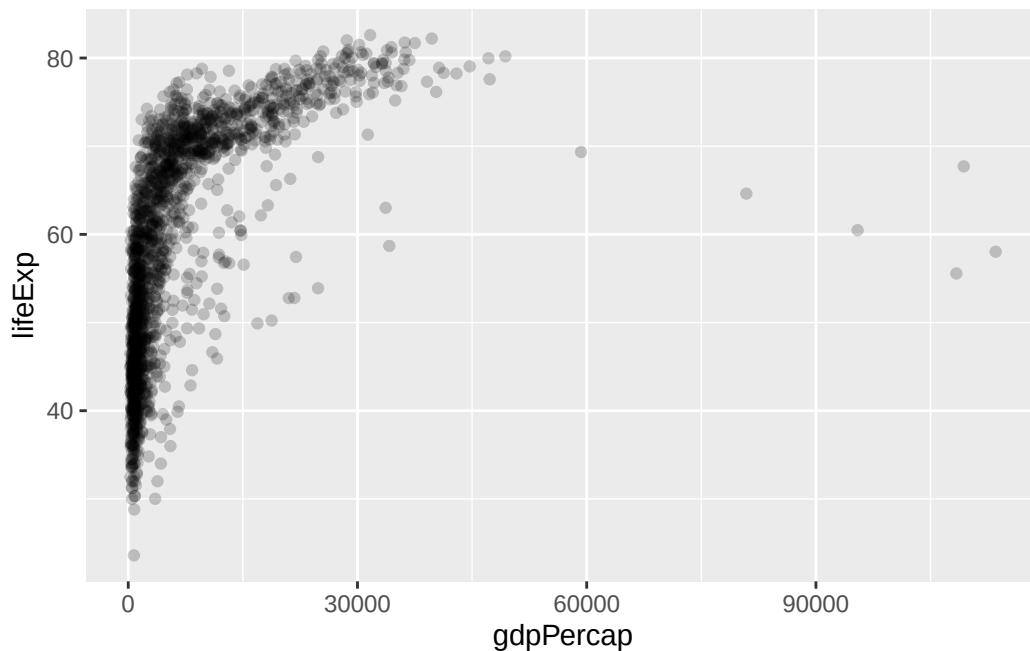
The following objects are masked from 'package:stats':

filter, lag

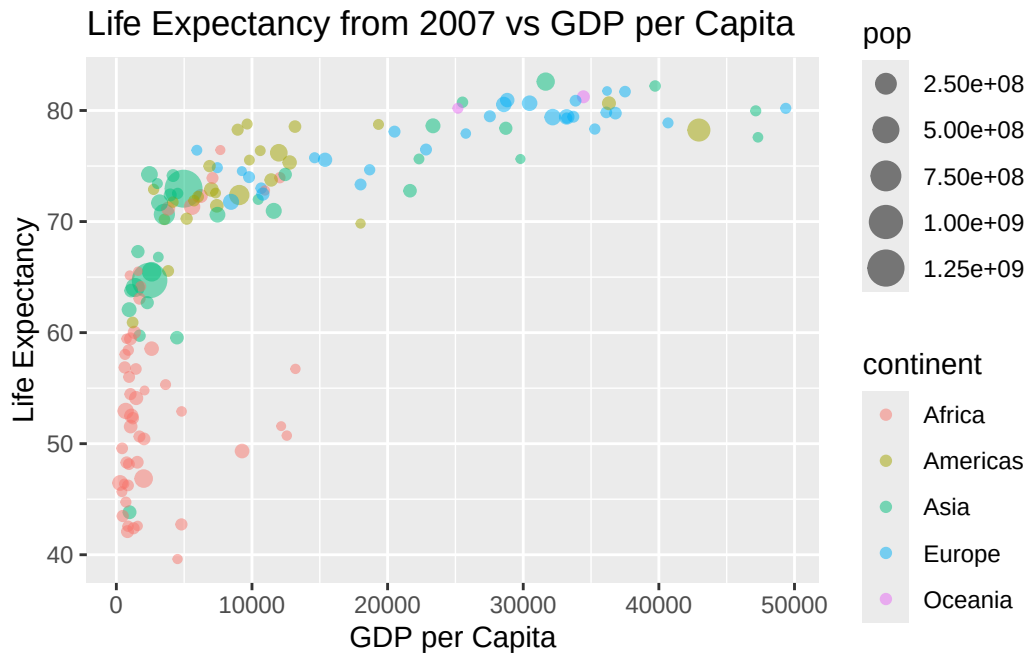
The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
ggplot(gapminder) +  
  aes(x=gdpPercap, y=lifeExp) +  
  geom_point(alpha=0.2)
```

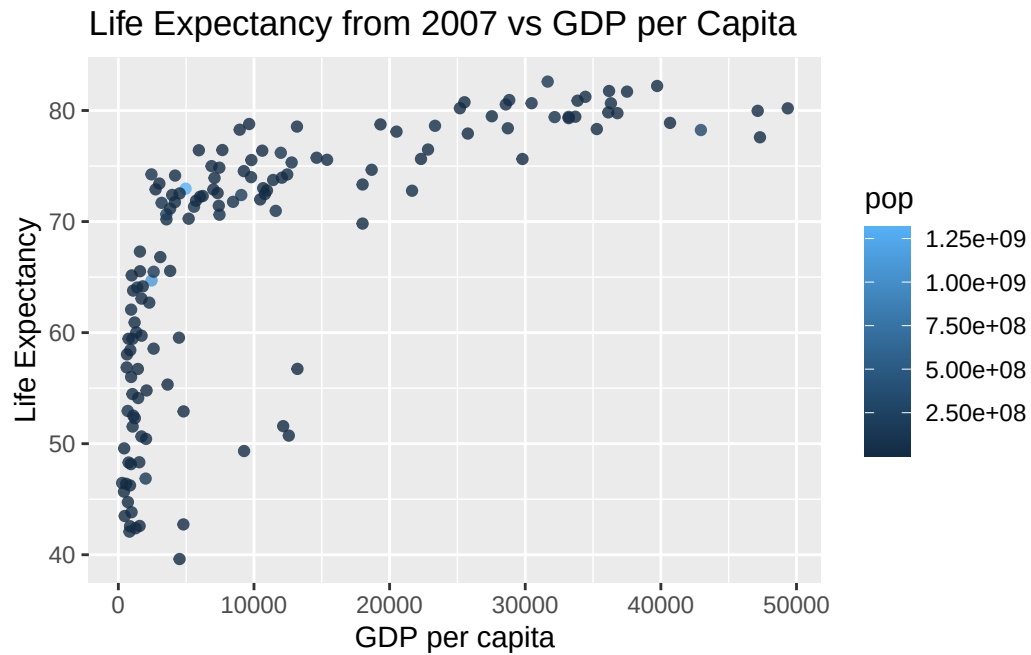


```
library(dplyr)  
gapminder_2007 <- gapminder %>% filter(year==2007)  
ggplot(gapminder_2007) +  
  aes(x=gdpPercap, y=lifeExp, color=continent, size=pop) +  
  geom_point(alpha=0.5) +  
  labs (title = "Life Expectancy from 2007 vs GDP per Capita",  
        x= "GDP per Capita",  
        y = "Life Expectancy")
```



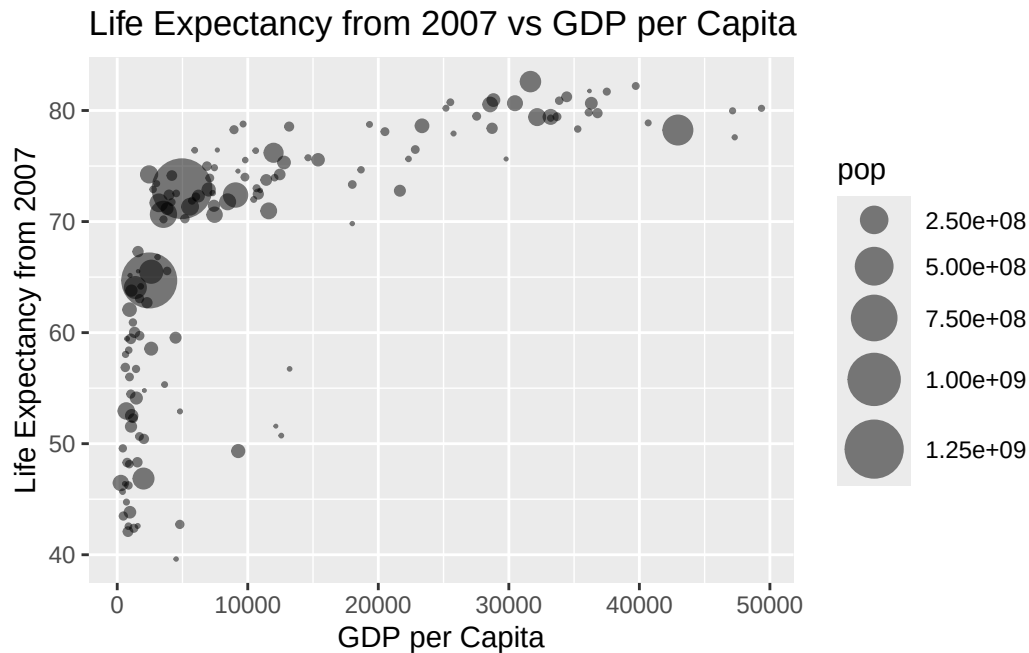
If we colored the points by the numeric variable population:

```
ggplot(gapminder_2007) +
  aes(x = gdpPercap, y = lifeExp, color = pop) +
  geom_point(alpha=0.8) +
  labs (title = "Life Expectancy from 2007 vs GDP per Capita",
        x= "GDP per capita",
        y = "Life Expectancy")
```



Section 7: Adjusting Point Size

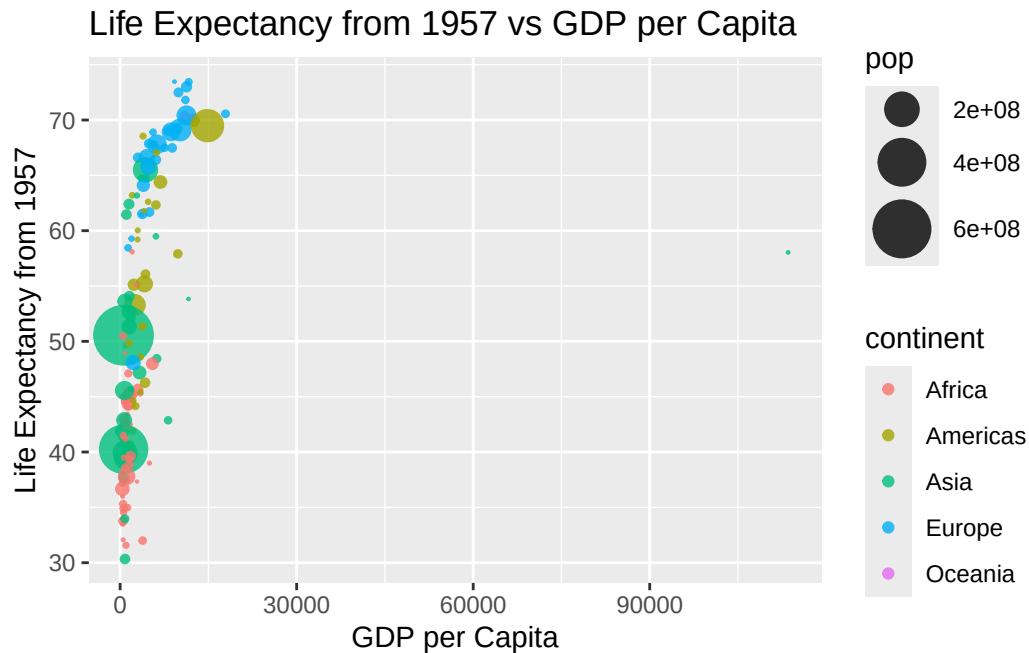
```
ggplot(gapminder_2007) +  
  aes(x = gdpPercap, y = lifeExp, size = pop) +  
  geom_point(alpha=0.5) +  
  scale_size_area(max_size = 10) +  
  labs (title = "Life Expectancy from 2007 vs GDP per Capita",  
        x= "GDP per Capita",  
        y = "Life Expectancy from 2007")
```



Q. Can you adapt the code you have learned thus far to reproduce our gapminder scatter plot for the year 1957? What do you notice about this plot is it easy to compare with the one for 2007?

```
gapminder_1957 <- gapminder %>% filter(year==1957)

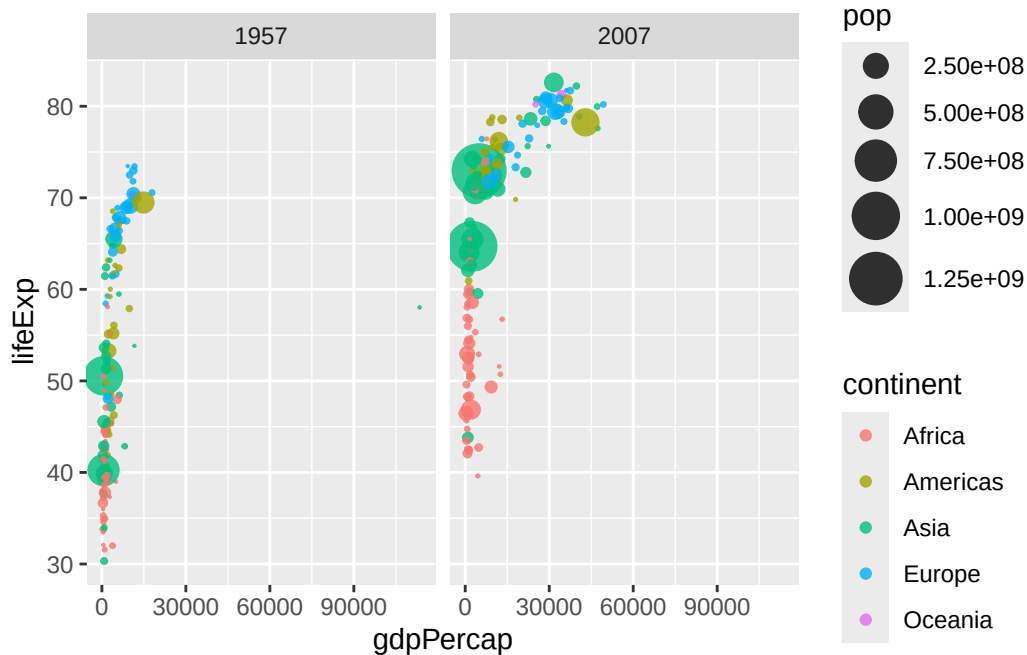
ggplot(gapminder_1957) +
  aes(x = gdpPerCap, y = lifeExp, color=continent,
      size = pop) +
  geom_point(alpha=0.8) +
  scale_size_area(max_size = 10) +
  labs (title = "Life Expectancy from 1957 vs GDP per Capita",
        x= "GDP per Capita",
        y = "Life Expectancy from 1957")
```



Q. Do the same steps above but include 1957 and 2007 in your input dataset for `ggplot()`. You should now include the layer `facet_wrap(~year)` to produce the following plot:

```
gapminder_1957_and_2007 <- gapminder %>% filter(year==1957 | year==2007)

ggplot(gapminder_1957_and_2007) +
  geom_point(aes(x = gdpPercap, y = lifeExp, color=continent,
                 size = pop), alpha=0.8) +
  scale_size_area(max_size = 9) +
  facet_wrap(~year)
```



10. Combining Plots

```
#install.packages("patchwork")
library(patchwork)
scatter <- ggplot(mtcars) + geom_point(aes(mpg, disp))
boxwhisker <- ggplot(mtcars) + geom_boxplot(aes(gear, disp, group = gear))
curve <- ggplot(mtcars) + geom_smooth(aes(dis, qsec))
bargraph <- ggplot(mtcars) + geom_bar(aes(carb))

#Using patchwork to combine them
(scatter | boxwhisker | curve) /
  bargraph
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

